

# Cancer Genetics & Hereditary Syndromes

Germline Testing, Hereditary Cancer Syndromes, and Risk Management

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## 1. Principles of Hereditary Cancer

Approximately 5-10% of all cancers arise in the setting of an inherited predisposition syndrome. These are characterised by high-penetrance germline pathogenic variants (PVs) in cancer susceptibility genes (CSGs), autosomal dominant (or less commonly recessive) inheritance, earlier age of onset, bilateral/multifocal disease, and characteristic tumour spectra. Identification enables risk-reduction strategies for the proband and cascade testing for at-risk relatives.

## 2. Major Hereditary Cancer Syndromes

| Syndrome             | Gene(s)                | Key Associated Cancers                                  | Surveillance Recommendations                        |
|----------------------|------------------------|---------------------------------------------------------|-----------------------------------------------------|
| HBOC                 | BRCA1 / BRCA2          | Breast (70-85%), Ovarian (17-46%), Pancreatic, Prostate | Annual MRI + mammo from age 25-30; consider RRBSO   |
| Lynch Syndrome       | MLH1, MSH2, MSH6, PMS2 | CRC, Endometrial, Ovarian, Gastric, Urological          | Annual colonoscopy from 20-25; endometrial sampling |
| FAP / AFAP           | APC                    | Near 100% CRC lifetime risk; gastric polyps             | Sigmoidoscopy from age 10-12; colectomy timing      |
| Li-Fraumeni Syndrome | TP53                   | Sarcoma, Brain, Breast (premenop), Adrenocortical       | Annual whole-body MRI; avoid radiation exposure     |

|                 |      |                                                     |                                                       |
|-----------------|------|-----------------------------------------------------|-------------------------------------------------------|
| Cowden Syndrome | PTEN | Breast, Thyroid, Endometrial, CRC                   | Annual breast MRI + mammo; thyroid US; colonoscopy    |
| MEN1            | MEN1 | Parathyroid, pituitary, pancreatic NETs             | Annual labs + imaging from age 5                      |
| MEN2A/2B        | RET  | MTC, pheochromocytoma, hyperparathyroidism          | Prophylactic thyroidectomy (age-dependent on variant) |
| VHL             | VHL  | Clear cell RCC, haemangioblastoma, pheochromocytoma | Annual MRI brain/spine/abdomen; ophthalmology         |

### 3. Germline Testing — Methodology & Indications

- Multi-gene panel testing (MGPT) via NGS — simultaneous analysis of 25-100+ CSGs; first-line approach for most cancer genetics referrals
- Single gene or 2-gene testing — reserved for specific phenotypes (e.g., BRCA1/2 only for Ashkenazi Jewish founder variant screening)
- Tumour-normal sequencing — somatic NGS can reveal germline PVs (e.g., BRCA1/2 in ovarian cancer); confirmatory germline testing required
- NCCN criteria for testing:  $\geq 3$  relatives with Lynch-associated cancers; breast cancer  $\leq 50$  years; triple-negative breast cancer  $\leq 60$  years;  $\geq 2$  primaries in one individual; specific ethnic risk (Ashkenazi Jewish)

### 4. Variant Classification (ACMG/AMP Framework)

| Class | Terminology                             | Clinical Action                                                     |
|-------|-----------------------------------------|---------------------------------------------------------------------|
| 5     | Pathogenic (PV)                         | Treat as disease-causing; offer risk management and cascade testing |
| 4     | Likely Pathogenic (LPV)                 | Manage as pathogenic per current evidence; reassess periodically    |
| 3     | Variant of Uncertain Significance (VUS) | Do NOT use clinically; re-classify as evidence accrues              |
| 2     | Likely Benign                           | Generally no clinical action                                        |
| 1     | Benign                                  | No clinical action                                                  |

■ **VUS HANDLING:** Variants of Uncertain Significance (VUS) must NOT be used to drive clinical management decisions. Patients and families should be counselled on the uncertainty. Laboratories reclassify VUS as evidence accrues — re-contact protocols are essential.

### 5. Risk Reduction Strategies — BRCA1/2 Example

- Risk-reducing salpingo-oophorectomy (RRSO): Reduces ovarian cancer risk by ~80-96%; recommended BRCA1 carriers age 35-40, BRCA2 age 40-45 after childbearing complete
- Risk-reducing mastectomy (RRM): Reduces breast cancer risk by >90%; individualized decision balancing risk, surveillance quality, and patient preference
- Chemoprevention: Tamoxifen, raloxifene, or aromatase inhibitors reduce breast cancer risk ~30-50% in high-risk women; oophorectomy is more effective
- Enhanced surveillance: Annual MRI + mammography from age 25-30 for BRCA1; 25-35 for BRCA2 (evidence supports MRI sensitivity ~75-80% vs mammography ~40%)